

L3ak CTF 2025 03 Walkthrough

Wi-Fight A Ghost?

Scenario

A **Ghost** never stays in one place for long. We have been hunting one and intercepted his device after a recent op. We believe it holds clues to his movements across the city.

Your task is to retrace his footsteps. Analyze system artifacts, browser history, and wireless configurations.

Find out:

- Which networks he connected to
- When and where he connected
- What he was doing while online

This isn't just digital — his path through the physical world is buried in the data.

Initial Look

We are given **KAPE** output containing Windows artifacts.

```
(kali㉿kali) - [~/Desktop/KAPEOUT/C]
$ tree -la -L 2
.
├── $Boot
├── $Extend
│   ├── $J
│   ├── $Max
│   └── $RmMetadata
├── $LogFile
├── $MFT
├── $Secure_$SDS
├── ProgramData
│   ├── Bromium
│   ├── Hewlett-Packard
│   ├── HP
│   ├── HPNotification_setup.log
│   ├── Intel
│   ├── Microsoft
│   ├── Microsoft OneDrive
│   ├── ntuser.pol
│   ├── Packages
│   ├── regid.1991-06.com.microsoft
│   ├── USOPrivate
│   ├── USOShared
│   └── win-add.mof
├── Users
│   ├── Default
│   ├── NotVi
│   └── Public
└── Windows
    ├── appcompat
    ├── inf
    ├── prefetch
    ├── ServiceProfiles
    └── System32

24 directories, 9 files
```

Most of the questions can be answered using the Registry files found in `config`. Many answers are contained in the logs from `winevent` as well.

```
(kali㉿kali) - [~/.../KAPEOUT/C/Windows/System32]
$ tree -la -L 1
.
├── config
├── grouppolicy
├── LogFiles
├── SleepStudy
├── SRU
├── Tasks
├── wbem
├── WDI
└── winevt

10 directories, 0 files
```

Regripper will make the tasks easier.

Questions

1. What was the hostname of the device?

Using regripper on SYSTEM.

```
Launching compname v.20090727
compname v.20090727
(System) Gets ComputerName and Hostname values from System hive

ComputerName      = 99PHOENIXDOWNS
TCP/IP Hostname   = 99PhoenixDowns
```

Answer: 99PHOENIXDOWNS

2. What was the SSID of the first Wi-Fi network they connected to?

Using regripper on SYSTEM

```

Adapter: {18c11dbd-93ab-4ca9-a804-4f4475da25b8}/D6577637F57657563747F55374
SSID Decoded: mugs_guest_5G
LastWrite Time: 2025-05-14 00:35:07Z
  EnableDHCP                1
  Domain
  NameServer
  DhcpIPAddress              192.168.0.114
  DhcpSubnetMask              255.255.255.0
  DhcpServer                  192.168.0.1
  Lease                       7200
  LeaseObtainedTime           2025-05-14 00:13:36Z
  T1                           2025-05-14 01:13:36Z
  T2                           2025-05-14 01:58:36Z
  LeaseTerminatesTime          2025-05-14 02:13:36Z
  AddressType                  0
  IsServerNapAware             0
  DhcpConnForceBroadcastFlag    0
  DhcpNetworkHint              mugs_guest_5G
  DhcpDefaultGateway            192.168.0.1
  DhcpNameServer                192.168.0.1
  DhcpSubnetMaskOpt             255.255.255.0
  DhcpInterfaceOptions          ü!y!w!/?!.,!+!!!! À"Q ỹ99PhoenixDowns À" À"ỹ ỹỹ; ;: 3 6 À"5
  DhcpGatewayHardware           À" T ñ
  DhcpGatewayHardwareCount      1

```

mugs_guest_5G

3. When did they obtain the DHCP lease at the first café?

2025-05-14 00:13:36

4. What IP address was assigned at the first café?

192.168.0.114

5. What GitHub page did they visit at the first café?

Internet history is typically stored in a **SQL** database in AppData . Looking in KAPEOUT/C/Users/NotVi/AppData/Local/Microsoft/Edge/User Data/Default can open History with **sqlitebrowser**. Looking at the **urls** table.

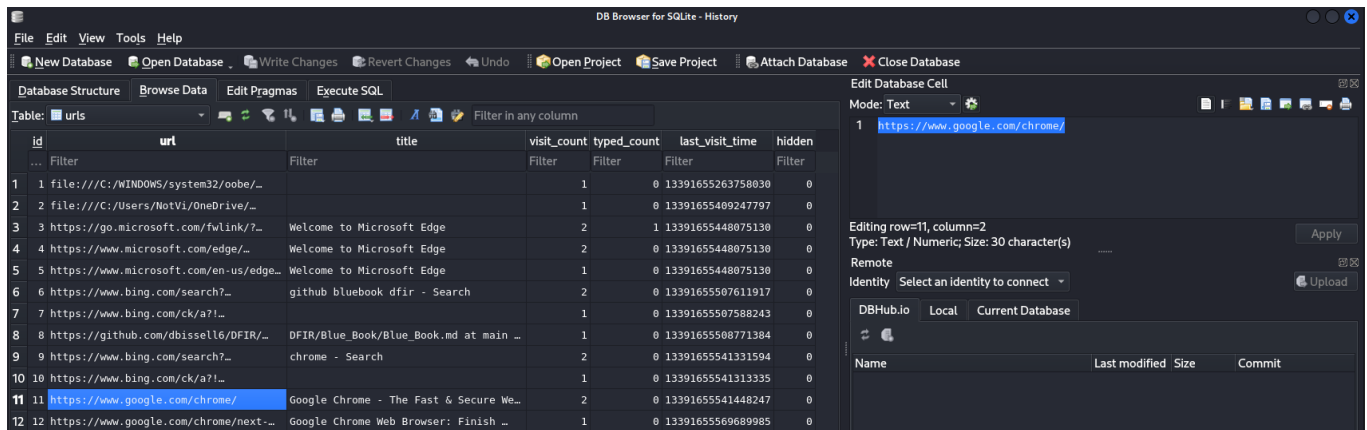
id	url	title	visit_count	typed_count	last_visit_time	hidden
1	file:///C:/WINDOWS/system32/oobe/...		1	0	13391655263758030	0
2	file:///C:/Users/NotVi/OneDrive/...		1	0	13391655409247797	0
3	https://go.microsoft.com/fwlink/?...	Welcome to Microsoft Edge	2	1	13391655448075130	0
4	https://www.microsoft.com/edge/...	Welcome to Microsoft Edge	2	0	13391655448075130	0
5	https://www.microsoft.com/en-us/edge...	Welcome to Microsoft Edge	1	0	13391655448075130	0
6	https://www.bing.com/search?...	github bluebook dfir - Search	2	0	13391655507611917	0
7	https://www.bing.com/ck/a?!...		1	0	13391655507588243	0
8	https://github.com/dbissell6/DFIR/...	DFIR/Blue_Book/Blue_Book.md at main ...	1	0	13391655508771384	0
9	https://www.bing.com/search?...	chrome - Search	2	0	13391655541331594	0
10	https://www.bing.com/ck/a?!...		1	0	13391655541313335	0
11	https://www.google.com/chrome/	Google Chrome - The Fast & Secure We...	2	0	13391655541448247	0
12	https://www.google.com/chrome/next-...	Google Chrome Web Browser: Finish ...	1	0	13391655569689985	0

They visited a **GitHub** page on DFIR.

https://github.com/dbissell6/DFIR/blob/main/Blue_Book/Blue_Book.md

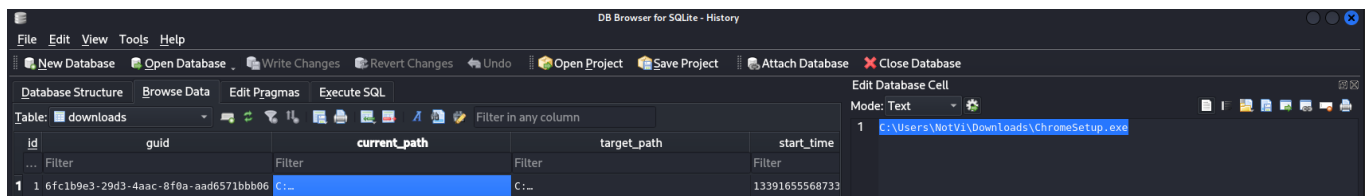
6. What file did they download at the first café?

Further down on the same table we see they went to download **Chrome**.



id	url	title	visit_count	typed_count	last_visit_time	hidden
1	file:///C:/WINDOWS/system32/oobe/...		1	0	13391655263758030	0
2	file:///C:/Users/NotVi/OneDrive/...		1	0	13391655409247797	0
3	https://go.microsoft.com/fwlink/?...	Welcome to Microsoft Edge	2	1	13391655448075130	0
4	https://www.microsoft.com/edge/...	Welcome to Microsoft Edge	2	0	13391655448075130	0
5	https://www.microsoft.com/en-us/edge...	Welcome to Microsoft Edge	1	0	13391655448075130	0
6	https://www.bing.com/search?...	github bluebook dfir - Search	2	0	13391655507611917	0
7	https://www.bing.com/ck/a?!...		1	0	13391655507588243	0
8	https://github.com/dbissell6/DFIR/...	DFIR/Blue_Book/Blue_Book.md at main ...	1	0	13391655508771384	0
9	https://www.bing.com/search?...	chrome - Search	2	0	13391655541331594	0
10	https://www.bing.com/ck/a?!...		1	0	13391655541313355	0
11	https://www.google.com/chrome/	Google Chrome - The Fast & Secure We...	2	0	13391655541448247	0
12	https://www.google.com/chrome/next...	Google Chrome Web Browser: Finish ...	1	0	13391655569689985	0

Shifting to the **downloads** table in the db.



id	guid	current_path	target_path	start_time
1	6fc1b9e3-29d3-4aac-8f8a-aad6571bbb06	C:\...	C:\...	13391655568733

In **regripper** **SYSTEM** we can see further evidence of the file.

```
-----
Launching bam v.20200427
bam v.20200427
(System) Parse files from System hive BAM Services

S-1-5-21-1894728654-1149273517-574531550-1000
2025-05-14 00:13:59Z - \Device\HarddiskVolume3\Windows\explorer.exe
2025-05-14 00:13:59Z - Microsoft.Windows.CloudExperienceHost_cw5nlh2txyewy
2025-05-13 23:30:43Z - Microsoft.Windows.00BENetworkConnectionFlow_cw5nlh2txyewy
2025-05-13 23:43:28Z - \Device\HarddiskVolume3\Windows\System32\CredentialUIBroker.exe

S-1-5-21-1894728654-1149273517-574531550-1001
2025-05-14 00:14:21Z - \Device\HarddiskVolume3\Windows\explorer.exe
2025-05-14 00:15:36Z - \Device\HarddiskVolume3\Windows\System32\oobe\FirstLogonAnim.exe
2025-05-14 02:14:39Z - Microsoft.Windows.Client.CBS_cw5nlh2txyewy
2025-05-14 00:16:38Z - Microsoft.Windows.StartMenuExperienceHost_cw5nlh2txyewy
2025-05-14 00:34:44Z - \Device\HarddiskVolume3\Windows\System32\cmd.exe
2025-05-14 00:39:09Z - Microsoft.Windows.Notepad_8wekyb3d8bbwe
2025-05-14 01:05:27Z - AD2F1837.myHP_v10z8vjag6ke6
2025-05-14 00:55:29Z - \Device\HarddiskVolume3\Program Files\HP\Sure Click\servers\BrConsole.exe
2025-05-14 00:50:25Z - \Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
2025-05-14 00:20:23Z - \Device\HarddiskVolume3\Users\NotVi\Downloads\ChromeSetup.exe
2025-05-14 00:20:23Z - \Device\HarddiskVolume3\Users\NotVi\AppData\Local\Temp\Google960_2086189827\bin\updater.exe
2025-05-14 02:19:01Z - \Device\HarddiskVolume3\Program Files\Google\Chrome\Application\chrome.exe
```

ChromeSetup.exe

7. They started taking notes — what was the name of the file?

From **regripper -r NTUSER.DAT -a @ KAPEOUT/C/Users/NotVi/** we can view RecentDocs. This shows a text file.

```

-----
Launching recentdocs v.20200427
recentdocs v.20200427
(NTUSER.DAT) Gets contents of user's RecentDocs key

RecentDocs
**All values printed in MRUList\MRUListEx order.
Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs
LastWrite Time: 2025-05-14 00:44:11Z
3 = The Internet
2 = &suppressAnimations=false&showFooter=true&allowPageNavigation=true&edgeGestureOffset=0&inputAnimationSourceId=0&inputAnimationProviderId=0
1 = Desktop
6 = D:\
5 = SAMDATA (D:)
4 = KAPEOUT
0 = HowToHackTheWorld.txt

Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\.txt
LastWrite Time 2025-05-14 00:16:49Z
MRUListEx = 0
0 = HowToHackTheWorld.txt

```

HowToHackTheWorld.txt

8. What is the contents of the notes?

Looking at the .lnk to the file we can see it it should be on the onedrive desktop

```

(kali@kali) - [~/.../Roaming/Microsoft/Windows/Recent]
$ exiftool HowToHackTheWorld.lnk
ExifTool Version Number      : 13.10
File Name                    : HowToHackTheWorld.lnk
Directory                    : .
File Size                    : 550 bytes
File Modification Date/Time   : 2025:05:13 18:16:49-06:00
File Access Date/Time        : 2025:05:15 21:19:38-06:00
File Inode Change Date/Time   : 2025:05:13 22:37:24-06:00
File Permissions              : -rwxr-xr-x
File Type                    : LNK
File Type Extension           : lnk
MIME Type                    : application/octet-stream
Flags                        : IDList, LinkInfo, RelativePath, WorkingDir, Unicode, NoKnownFolderTracking
File Attributes               : (none)
Target File Size              : 0
Icon Index                   : (none)
Run Window                   : Normal
Hot Key                      : (none)
Target File DOS Name          : HowToHackTheWorld.txt
Drive Type                   : Fixed Disk
Drive Serial Number           : 30E5-4C6D
Volume Label                  : Windows
Local Base Path               : C:\Users\NotVi\OneDrive\Desktop\HowToHackTheWorld.txt
Relative Path                 : ..\..\..\..\OneDrive\Desktop\HowToHackTheWorld.txt
Working Directory              : C:\Users\NotVi\OneDrive\Desktop

```

Going to the KAPE output we dont have that file.

```

(kali@kali) - [~/.../KAPEOUT/C/Users/NotVi]
$ tree -la -L 2
.
├── AppData
│   ├── Local
│   └── Roaming
├── NTUSER.DAT
├── NTUSER.DAT{07dcf1db-3062-11f0-93ab-815d75251871}.TM.blf
├── NTUSER.DAT{07dcf1db-3062-11f0-93ab-815d75251871}.TMContainer000000000000000001.regtrans-ms
├── ntuser.dat.LOG1
└── ntuser.dat.LOG2

4 directories, 5 files

```

Looking at the Orange box we can see the file is really small. If the file is smaller than ~900 bytes it could be stored in the MFT itself.

Using **Chainsaw** to dump and decode the MFT.

```
(kali@kali) - [~/Desktop/KAPEOUT/C]
$ ~/Tools/chainsaw/chainsaw dump \$MFT --output mft.output --decode-data-streams

CHAINS AW
By WithSecure Countercept (@FranticTyping, @AlexKornitzer)

[+] Dumping the contents of forensic artefacts from: $MFT (extensions: *)
[+] Loaded 1 forensic artefacts (861.9 MB)
```

```
---
Signature: FILE
EntryId: 79305
Sequence: 9
BaseEntryId: 0
BaseEntrySequence: 0
HardLinkCount: 2
Flags: ALLOCATED
UsedEntrySize: 904
TotalEntrySize: 1024
FileSize: 29
IsADirectory: false
IsDeleted: false
HasAlternateDataStreams: false
StandardInfoFlags: FILE_ATTRIBUTE_ARCHIVE | FILE_ATTRIBUTE_REPARSE_POINT
StandardInfoLastModified: 2025-05-14T00:16:49.224155Z
StandardInfoLastAccess: 2025-05-14T00:35:44.083666Z
StandardInfoCreated: 2025-05-14T00:16:49.107063Z
FileNameFlags: FILE_ATTRIBUTE_ARCHIVE
FileNameLastModified: 2025-05-14T00:16:49.224155Z
FileNameLastAccess: 2025-05-14T00:16:49.224155Z
FileNameCreated: 2025-05-14T00:16:49.107063Z
FullPath: Users/NotVi/OneDrive/Desktop/HowToHackTheWorld.txt
DataStreams:
- stream_name: ''
  stream_number: 0
  stream_data: Practice and take good notes.
---
```

Practice and take good notes

9. What was the SSID of the second Wi-Fi network they connected to?

In **regripper** SYSTEM

```

Adapter: {18c11dbd-93ab-4ca9-a804-4f4475da25b8}
LastWrite Time: 2025-05-14 00:35:07Z
  EnabledDHCP                1
  Domain
  NameServer
  DhcpIPAddress              10.0.6.28
  DhcpSubnetMask             255.255.240.0
  DhcpServer                 10.0.0.1
  Lease                      14400
  LeaseObtainedTime          2025-05-14 00:35:07Z
  T1                         2025-05-14 02:35:07Z
  T2                         2025-05-14 04:05:07Z
  LeaseTerminatesTime        2025-05-14 04:35:07Z
  AddressType                0
  IsServerNapAware           0
  DhcpConnForceBroadcastFlag 0
  DhcpNetworkHint            AlleyCat
  DhcpInterfaceOptions       ü=!y=!w=!/=!.=!,=!+=!!!=!8@
Q"8@ÿÿ99PhoenixDowns.myrtle.solutions8@myrtle.solutions8@ÿÿö;8@18:8@ 38@8@68@
58@
  DhcpDefaultGateway         10.0.0.1
  DhcpNameServer              8.8.8.8 8.8.4.4
  DhcpDomain                  myrtle.solutions
  DhcpSubnetMaskOpt           255.255.240.0
  DhcpGatewayHardware
@Cn·8
  DhcpGatewayHardwareCount    1

```

AlleyCat

10. When did they obtain the second lease?

2025-05-14 00:35:07

11. What was the IP address assigned to the device at the second café?

10.0.6.28

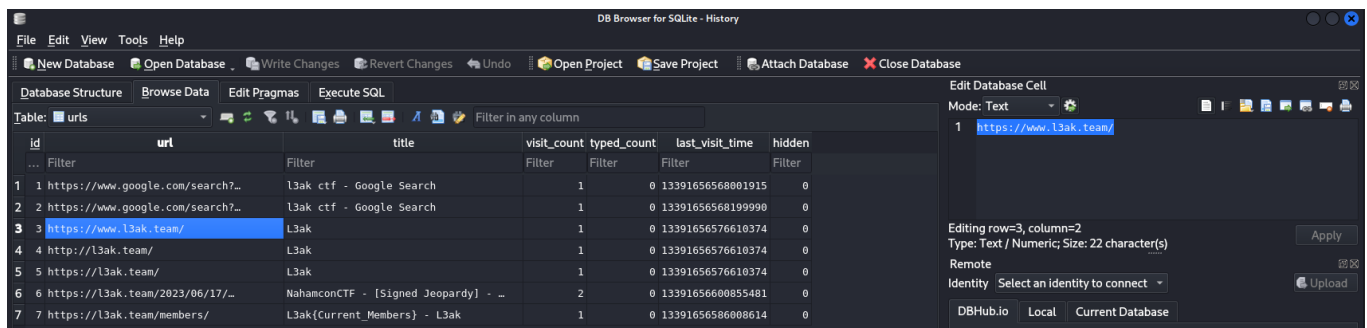
12. What hostname appeared in the DHCP options at the second café?

99PhoenixDowns.myrtle.solutions ****

13. What website did they log into at the second café?

From the first couple questions we found that they downloaded **Chrome**. This information can be accessed like before at `KAPEOUT/C/Users/NotVi/AppData/Local/Google/Chrome/User Data/Default`

Again, using **sqlitebrowser** to view `History`. Looking at the `url` table the answer is in the 3rd row.



`https://www.l3ak.team`

14. What was the MAC address of the Wi-Fi adapter used at the cafés?

In `regripper` looking at `SYSTEM` we can see the `Wi-Fi PermanentAddress`

```
Launching networksetup2 v.20191004
networksetup2 v.20191004
(System) Get NetworkSetup2 subkey info

Wi-Fi - Intel(R) Wireless-AC 9560 160MHz (wireless)
  CurrentAddress   : 48:51:c5:35:ea:53
  PermanentAddress : 48:51:c5:35:ea:53
Bluetooth Network Connection - Bluetooth Device (Personal Area Network) (wired)
  CurrentAddress   : 48:51:c5:35:ea:57
  PermanentAddress : 48:51:c5:35:ea:57
Local Area Connection* 3 - Microsoft Wi-Fi Direct Virtual Adapter #3 (wireless)
  CurrentAddress   : 48:51:c5:35:ea:54
  PermanentAddress : 48:51:c5:35:ea:54
Local Area Connection* 4 - Microsoft Wi-Fi Direct Virtual Adapter #4 (wireless)
  CurrentAddress   : 4a:51:c5:35:ea:53
  PermanentAddress : 4a:51:c5:35:ea:53
Ethernet - Realtek PCIe GbE Family Controller (wired)
  CurrentAddress   : c8:5a:cf:ba:b8:38
  PermanentAddress : c8:5a:cf:ba:b8:38
-----
```

We can confirm this using the logs from `winevt` .

```
---
Event_attributes:
  xmlns: http://schemas.microsoft.com/win/2004/08/events/event
Event:
  System:
    Provider_attributes:
      Name: Microsoft-Windows-Dhcp-Client
      Guid: 15A7A4F8-0072-4EAB-ABAD-F98A4D666AED
    EventID: 50065
    Version: 0
    Level: 4
    Task: 3
    Opcode: 55
    Keywords: '0x4000000000000000'
    TimeCreated_attributes:
      SystemTime: 2025-05-13T23:35:24.848488Z
    EventRecordID: 6
    Correlation: null
    Execution_attributes:
      ProcessID: 2076
      ThreadID: 2368
    Channel: Microsoft-Windows-Dhcp-Client/Admin
    Computer: 99PhoenixDowns
    Security_attributes:
      UserID: S-1-5-19
  EventData:
    NetworkHintString: mugs_guest_5G
    NetworkHint: D6577637F57657563747F55374
    HWLength: 6
    HWAddress: 4851C535EA53
---
```

Looking again at the second café.

```
---
Event_attributes:
  xmlns: http://schemas.microsoft.com/win/2004/08/events/event
Event:
  System:
    Provider_attributes:
      Name: Microsoft-Windows-Dhcp-Client
      Guid: 15A7A4F8-0072-4EAB-ABAD-F98A4D666AED
    EventID: 50067
    Version: 0
    Level: 4
    Task: 3
    Opcode: 57
    Keywords: '0x4000000000000000'
    TimeCreated_attributes:
      SystemTime: 2025-05-14T00:35:07.501564Z
    EventRecordID: 11
    Correlation: null
    Execution_attributes:
      ProcessID: 2052
      ThreadID: 3052
    Channel: Microsoft-Windows-Dhcp-Client/Admin
    Computer: 99PhoenixDowns
    Security_attributes:
      UserID: S-1-5-19
  EventData:
    NetworkHintString: AlleyCat
    NetworkHint: 14C6C65697341647
    HWLength: 6
    HWAddress: 4851C535EA53
```

48:51:c5:35:ea:53

15. What was the Wi-Fi password of the first café?

```
(kali㉿kali) - [~/Wlansvc/Profiles/Interfaces/{18C11DBD-93AB-4CA9-A804-4F4475DA25B8}]
$ strings *
<?xml version="1.0"?>
<WLANProfile xmlns="http://www.microsoft.com/networking/WLAN/profile/v1">
  <name>mugs_guest_5G</name>
  <SSIDConfig>
    <SSID>
      <hex>6D7567735F67756573745F3547</hex>
      <name>mugs_guest_5G</name>
    </SSID>
  </SSIDConfig>
  <connectionType>ESS</connectionType>
  <connectionMode>auto</connectionMode>
  <MSM>
    <security>
      <authEncryption>
        <authentication>WPA3SAE</authentication>
        <encryption>AES</encryption>
        <useOneX>false</useOneX>
        <transitionMode xmlns="http://www.microsoft.com/networking/WLAN/profile/v4">true</transitionMode>
      </authEncryption>
      <sharedKey>
        <keyType>passPhrase</keyType>
        <protected>true</protected>
        <keyMaterial>01000000D08C9DDF0115D118C7A00C04FC297EB0100000027BA46BE60165F489902447DADDD37640000000
002000000000106600000001000020000000B71A769EE992B83996ACA0EA2DDBD36FB687258C5F465FE82B76A88162AFD18600000000E800000000200002000000
053332437AB2888DBDBE294DAA0FEB07172871F6D1C20F0EEB1A9C0F03A4440711000000070C4F37EE924D6D41E9FB19BF319753240000000F85ABD2CD15A0FDBE8B
A653A75ADEF9A2CCEAA6E9D61E34D4ACB7D29A072EEAE84905134174A5C59BAFFE042869B18D07D4B3AA9E4329945235F59A46F7B98</keyMaterial>
      </sharedKey>
    </security>
  </MSM>
  <MacRandomization xmlns="http://www.microsoft.com/networking/WLAN/profile/v3">
    <enableRandomization>false</enableRandomization>
    <randomizationSeed>2081371451</randomizationSeed>
  </MacRandomization>
</WLANProfile>
```

[Requires DPAPI decryption — not answered yet]

16. What city did all of this take place in?

Again using **regripper** on **SYSTEM** can see the device is in Mountain Time.

```
-----
Launching timezone v.20200518
timezone v.20200518
(System) Get TimeZoneInformation key contents

TimeZoneInformation key
ControlSet001\Control\TimeZoneInformation
LastWrite Time 2025-05-14 00:14:19Z
  DaylightName    -> Mountain Daylight Time
  StandardName    -> Mountain Standard Time
  Bias            -> 420 (7 hours)
  ActiveTimeBias  -> 360 (6 hours)
  TimeZoneKeyName-> Mountain Standard Time
-----
```


Fort Collins